24/10/2026

Code-A

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MM : 720 **PRE- AIATS-02(Class XI for NEET-2028)_ for ACHIEVERS** **Time : 180 Min.**

Topics Covered:
Physics: Motion in a Plane
Chemistry: Classification of Elements and Periodicity in Properties, Chemical Bonding and Molecular Structure
Botany: Cell Cycle & Cell Division, The Living World
Zoology: Breathing & Exchange of Gases, Body Fluids and Circulation

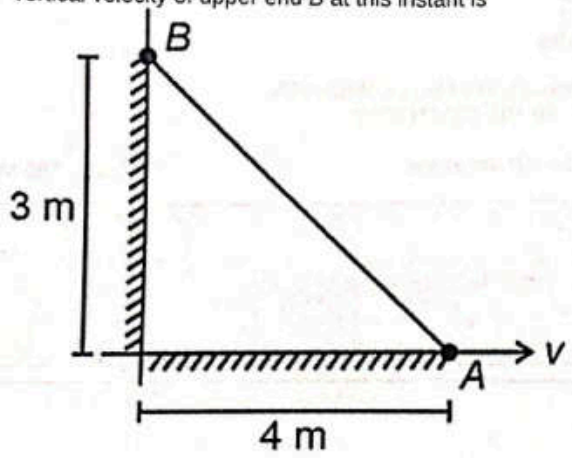
General Instructions :

1. Duration of Test is 3 hrs.
2. The Test consists of 180 questions. The maximum marks are 720.
3. There are four parts in the question paper consisting of **Physics**, **Chemistry**, **Botany** and **Zoology** having 45 questions in each part of equal weightage.
4. Each correct answer carries four marks. One mark will be deducted for each incorrect answer from the total score.
5. The clock will be set at the server. The countdown timer in the top right corner of screen will display the remaining time available for you to complete the examination. When the timer reaches zero, the examination will end by itself. You will not be required to end or submit your examination.
6. The Questions Palette displayed on the right side of screen will show the status of each question using one of the following:
   - Answered
   - Not Answered
   - Marked for Review
   - Not Visited
   - Answered and Marked for Review
7. Ensure that the battery in your device is sufficiently charged for serving 3 hr before starting the test. We recommend your device is on charging mode during the test.
8. Make sure you begin the test with a plan. Start with your strongest section.
9. Go through the entire paper and attempt the questions you know first.
10. Make sure you save at least 5-10 min in the end to revisit your answers. In an online test, you can change your answer at any time. You can only attempt the test from one device. You will be logged out from the first device if you log in from another device.
11. Don't change the date and time of the device in between the test.
12. Don't submit the test before time. Try to use the entire duration of the test wisely.

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Physics

1. A ladder rests against a smooth vertical wall as shown in figure. The floor is also smooth. Mass of ladder is 50 kg. If lower end A is moving with velocity v horizontally, then the vertical velocity of upper end B at this instant is



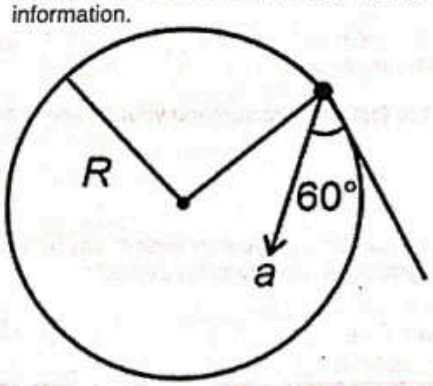
(1) v

(2) $2v$

(3) $\frac{3v}{4}$

(4) $\frac{4v}{3}$

2. In the given figure, $a = 20\ \text{m/s}^2$ represents the total acceleration of a particle moving in the clockwise direction in a circle of radius $R = 3\ \text{m}$ at a given instant of time. Choose the correct conclusion based upon the above information.



(1) The tangential acceleration of the particle is greater than centripetal acceleration

(2) The tangential acceleration of the particle is smaller than centripetal acceleration

(3) The tangential acceleration of the particle is equal to centripetal acceleration

(4) Both (1) and (3)

3. Two bullets are fired horizontally and simultaneously towards each other from roof top of two building 150 m apart and of same height of 200 m, with the velocity of 50 m/s and 25 m/s. The vertical distance from the ground where they will collide is

(1) 75 m

(2) 180 m

(3) 100 m

(4) 170 m

4. The speed of a swimmer in still water is 20 m/s. The speed of river water is 10 m/s and flowing due east. If he is standing on the south bank and wishes to cross the river in shortest possible time *i.e.*, 4 s, then the width of river is

(1) 50 m

(2) 100 m

(3) 110 m

(4) 80 m

5. A particle moves in a circle of radius 2 m with speed of particle varying as $2t^2$ where t is time in seconds. The acceleration of the particle at $t = 1$ s is

(1) $\sqrt{20}\ \text{m/s}^2$

(2) $\sqrt{40}\ \text{m/s}^2$

(3) $\sqrt{120}\ \text{m/s}^2$

(4) $\sqrt{205}\ \text{m/s}^2$

6. A ground to ground projectile is projected with speed v_0 at an angle θ with ground. The radius of curvature of trajectory of projectile at its initial point of projection is

(1) $\frac{v_0^2}{g}$

(2) $\frac{v_0^2}{g\sin\theta}$

(3) $\frac{v_0^2}{g\cos\theta}$

(4) $\frac{v_0^2}{g\tan\theta}$

7. Two balls of mass 5 kg and 15 kg are projected horizontally from the top of a tower of height 80 m simultaneously. Which ball will hit the ground first? (neglect air resistance)

(1) The ball with heavier mass

(2) The ball with lighter mass

(3) Both balls will reach simultaneously

(4) Cannot predict

8. If vectors, $\vec{A} = 2\hat{i} + 3\hat{j} + 6\hat{k}$ and $\vec{B} = 4\hat{i} + 5\hat{j} - 3\hat{k}$, then unit vector in the direction of $\vec{B} - \vec{A}$ is

(1) $\frac{2\hat{i}+2\hat{j}+9\hat{k}}{\sqrt{89}}$

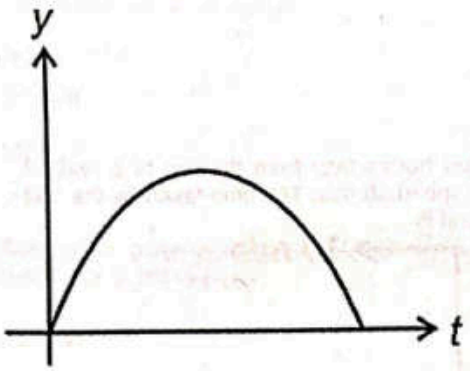
(2) $\frac{6\hat{i}+8\hat{j}-3\hat{k}}{\sqrt{78}}$

(3) $\frac{2\hat{i}+2\hat{j}-9\hat{k}}{\sqrt{89}}$

(4) $\frac{2\hat{i}+2\hat{j}-9\hat{k}}{\sqrt{78}}$

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9. Choose the correct conclusion regarding vertical component of velocity of a particle projected from ground whose position(y)-time(t) graph is as shown in the figure



(1) First increases then decreases

(2) First decreases then increases

(3) Increases continuously

(4) Remains constant

10. A person at rest observes that the rain is falling with 5 m/s towards east making an angle $53°$ with vertical. The velocity with which he should start walking so that rain starts appearing vertical is

(1) 3 m/s towards east

(2) 5 m/s towards east

(3) 4 m/s towards east

(4) 5 m/s towards north

11. For a given velocity of projection, the horizontal range of a ground to ground projectile is maximum when it is projected at an angle of

(1) $45°$

(2) $90°$

(3) $60°$

(4) $30°$

12. A particle started revolving from rest with constant angular acceleration α in a circular path of radius r. If tangential acceleration becomes numerically equal to twice the magnitude of centripetal acceleration in time t, then t is equal to

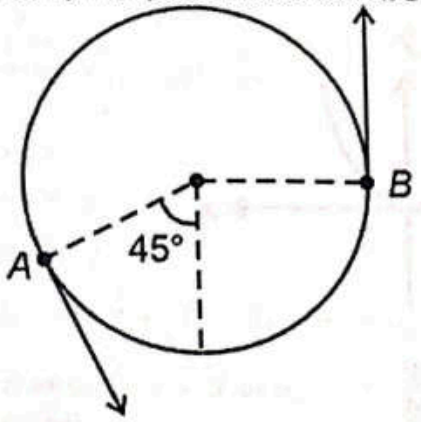
(1) $\frac{1}{\alpha}$

(2) $\sqrt{\frac{2}{\alpha}}$

(3) $\sqrt{\frac{1}{2\alpha}}$

(4) $\frac{1}{3\alpha}$

13. A particle of mass 5 kg is moving in circular path with constant speed of 30 m/s. The change in magnitude of velocity when particle travels from A to B will be



(1) $20\sqrt{3}$

(2) $30\sqrt{5}$

(3) Zero

(4) $10\sqrt{3}$

14. A swimmer can swim in still water at a speed of 4 m/s while the river is flowing at a speed of 5 m/s. If the swimmer wants to reach just opposite point of the bank then he should swim at an angle of

(1) $\sin^{-1}[3/5]$ with the vertical

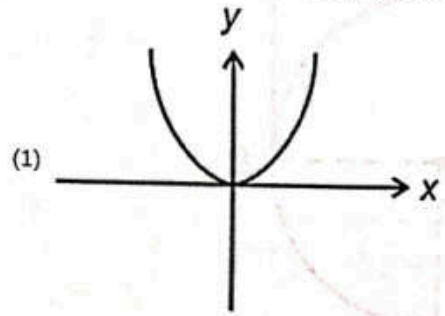
(2) $\sin^{-1}[4/5]$ with the horizontal

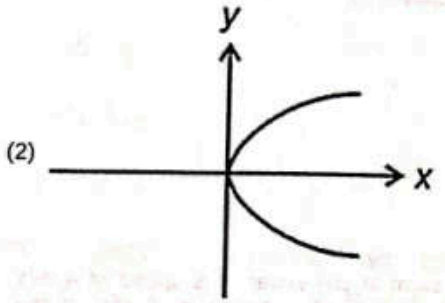
(3) Swimmer would never be able to reach opposite point

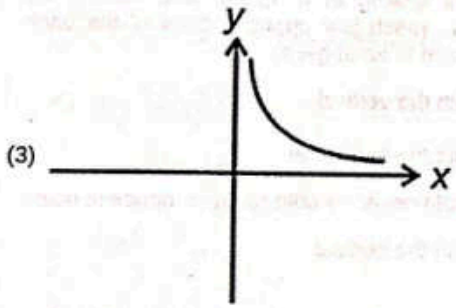
(4) $\sin^{-1}[3/4]$ with the vertical

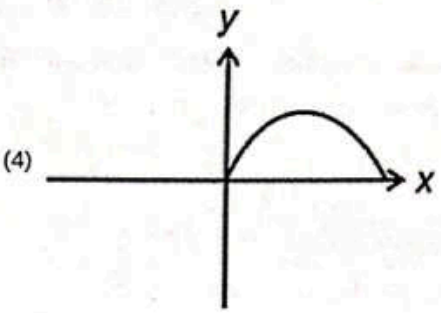
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15. The coordinates of a particle moving in *x-y* plane vary with time as per the following relation, $y = 5\sqrt{t}$ and $x = 2t$. The locus of the particle can be best represented by

(1) 

(2) 

(3) 

(4) 

16. A ball is projected with a velocity of $20\sqrt{3}$ m/s at an angle of 60° with horizontal. The time after which velocity vector will make an angle of 30° with the horizontal is (g = 10 m/s²)

(1) 3 s

(2) 5 s

(3) 4 s

(4) 2 s

17. An aeroplane flies 400 m south and then 300 m east and then it flies 1200 m downwards. The net displacement is

(1) 1400 m

(2) 1000 m

(3) 1300 m

(4) 1500 m

18. The angle between velocity and acceleration of a particle in a circular motion with increasing speed cannot be

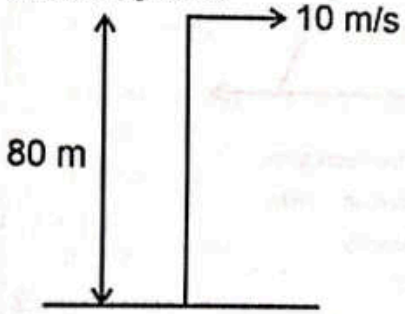
(1) 90°

(2) 45°

(3) 30°

(4) 60°

19. A ball is projected horizontally from the top of a wall of height 80 m with speed 10 m/s. The time taken by the ball to reach the ground is



(1) 2 second

(2) 5 second

(3) 3 second

(4) 4 second

20. Angular position of a body undergoing circular motion varies with time as $\theta = (4t + 3)$ radian. If radius of circular path is 1 m, then acceleration of body at t = 1 s will be

(1) 10 m/s²

(2) 16 m/s²

(3) 15 m/s²

(4) 5 m/s²

21. For angles of projection of a projectile at angles 90° + θ and 90° – θ, the maximum height attained by projectile are in ratio (Assume the velocity of projection of both projectiles to be same)

(1) 3 : 5

(2) 5 : 3

(3) 1 : 1

(4) 2 : 5

22. A cricket ball of mass m is hitted at an angle 30° with the horizontal, with velocity v_0. Its kinetic energy at topmost point is (use kinetic energy $K = \frac{1}{2}mv^2$)

(1) Zero

(2) $\frac{1}{2}mv_0^2$

(3) $\frac{3}{8}mv_0^2$

(4) $\frac{mv_0^2}{8}$

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23. A ship A is moving westwards with a speed of 30 km h^{-1} and a ship B 100 km south of A, is moving northwards with a speed 40 km h^{-1}. The time after which the distance between them becomes shortest is

(1) $10\sqrt{2}$ h

(2) 5 h

(3) $\frac{8}{5}$ h

(4) 2 h

24. Among the given options choose the one which does not represent a unit vector.

(1) $\frac{\hat{i}+\hat{j}}{\sqrt{2}}$

(2) $\frac{\hat{i}-\hat{j}}{\sqrt{2}}$

(3) $\frac{4\hat{i}-3\hat{j}}{5}$

(4) $\frac{4\hat{i}+3\hat{j}}{\sqrt{3}}$

25. A particle is moving with an acceleration of $\vec{a}=\left(2t^2\hat{i}+3t\hat{j}\right)m/s^2$. If at t = 0, its velocity is $\left(2\hat{i}+2\hat{j}\right)m/s$, then its velocity at time t = 3 s is

(1) $\left(5\hat{i}+\frac{27}{9}\hat{j}\right)$ m/s

(2) $\left(20\hat{i}+\frac{31}{2}\hat{j}\right)$ m/s

(3) $\left(20\hat{i}-\frac{31}{2}\hat{j}\right)$ m/s

(4) $\left(5\hat{i}-\frac{27}{9}\hat{j}\right)$ m/s

26. A particle moves so that its position vector is given by $\vec{r}=4\cos\left(\frac{\pi t}{5}\right)\hat{i}+4\sin\left(\frac{\pi t}{5}\right)\hat{j}$, where r is in metres and t is in seconds. The trajectory of the particle is

(1) Circular

(2) Parabolic

(3) Elliptical

(4) Hyperbolic

27. Angular speed of a particle is increased from 360 rpm to 420 rpm in 10 seconds. The average angular acceleration of particle is

(1) 5π rad/s^2

(2) Zero

(3) $\frac{\pi}{5}$ rad/s^2

(4) $\frac{3\pi}{2}$ rad/s^2

28. Two stones of masses m and $2m$ are whirled in horizontal circles, the heavier one in a radius $\frac{r}{2}$ and lighter one in radius r. The tangential speed of lighter one is twice the speed of the heavier one. The ratio of centripetal acceleration experienced by them is

(1) 1 : 3

(2) 2 : 1

(3) 5 : 3

(4) 4 : 7

29. If $\left|\vec{A}+\vec{B}\right|=\left|\vec{A}\right|=\left|\vec{B}\right|$, then angle between A and B will be ($\vec{A}$ and $\vec{B}$ are not a null vector)

(1) 90°

(2) 120°

(3) 0°

(4) 60°

30. A particle of mass 1 kg moves along a circle of radius $\frac{10}{\pi}$ m with a constant tangential acceleration. The speed at the end of second revolution after beginning of the motion is 20 m/s. The tangential acceleration of the particle is

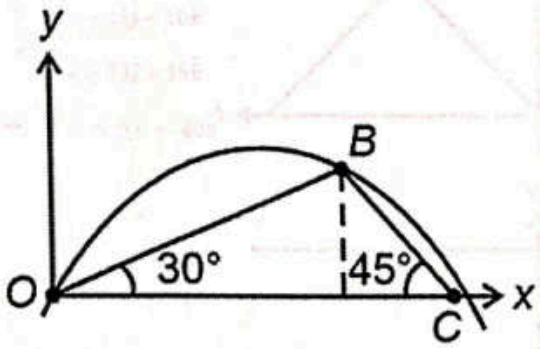
(1) 60π m/s^2

(2) $\frac{60}{\pi}$ m/s^2

(3) 5 m/s^2

(4) 30π m/s^2

31. A particle is projected from the ground at an angle θ with horizontal with some velocity. At any position B on the trajectory of particle the angle subtended by OB and BC with horizontal are 30° to 45° respectively. The angle of projection is



(1) $\tan^{-1}\left[\frac{\sqrt{3}-1}{2}\right]$

(2) $\tan^{-1}\left[\frac{\sqrt{3}+1}{\sqrt{3}}\right]$

(3) $\tan^{-1}\left[\frac{\sqrt{3}-1}{\sqrt{3}}\right]$

(4) $\tan^{-1}\left[\frac{\sqrt{3}+1}{\sqrt{5}}\right]$

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32. The path followed by a moving body whose magnitude of acceleration is constant and always acts in a direction perpendicular to velocity is

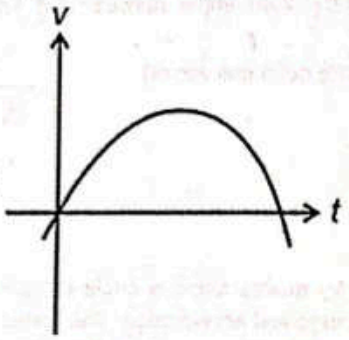
(1) Circular

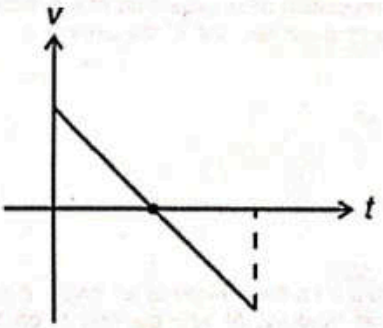
(2) Parabolic

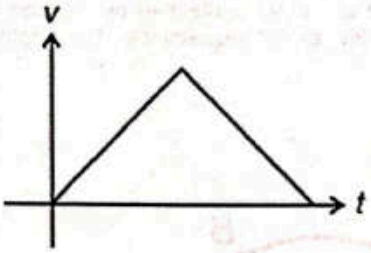
(3) Hellical

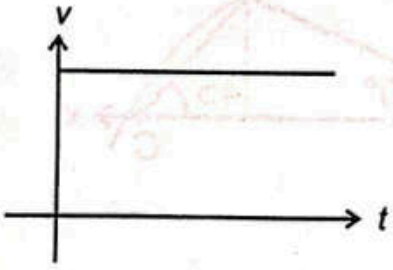
(4) Straight line

33. A particle is projected at an angle θ with horizontal from ground. The variation of horizontal component of velocity (v) of particle with time (t) is best represented by

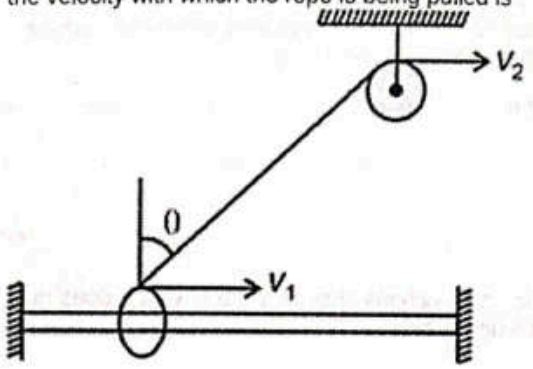
(1) 

(2) 

(3) 

(4) 

34. If the velocity of the ring is v_1 as shown in the figure then the velocity with which the rope is being pulled is



(1) $v_1 \cos\theta$

(2) $v_1 \sin\theta$

(3) $v_1 \tan\theta$

(4) $v_1 / \cos\theta$

35. A particle is projected with speed of 15 m/s at an angle 30° with horizontal. The range of the particle is

(1) $\frac{45\sqrt{3}}{4}$m

(2) $\frac{50\sqrt{3}}{3}$m

(3) $\frac{41\sqrt{3}}{2}$m

(4) $\frac{20\sqrt{5}}{3}$m

36. Horizontal range of a particle thrown from the ground is twice the maximum height. The angle of projection is

(1) 60°

(2) $\tan^{-1}(2)$

(3) 45°

(4) $\tan^{-1}(5)$

37. To a stationary man rain is falling on his back with a velocity v at an angle θ with horizontal. To make rain velocity appear vertical to the man, the man must move

(1) Forward with velocity $v\sin\theta$

(2) Backward with velocity $v\sin\theta$

(3) Forward with velocity $v\cos\theta$

(4) Backward with velocity $v\cos\theta$

38. A particle is projected with a velocity of 50 m/s at an angle of 53° with horizontal. The magnitude of position vector of particle, from point of projection, after 2 s will be

(1) 50 m

(2) $30\sqrt{3}m$

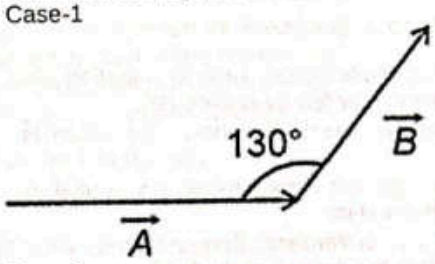
(3) $60\sqrt{2}m$

(4) $50\sqrt{2}m$

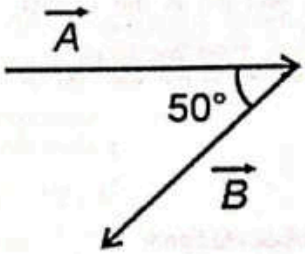
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39. The angle between the vectors $\vec{A}$ and $\vec{B}$ in two given cases respectively are

Case-1



Case-2



(1) 130°, 120°

(2) 50°, 50°

(3) 50°, 130°

(4) 130°, 50°

40. Choose the correct conclusions regarding the properties of vectors among the following options.

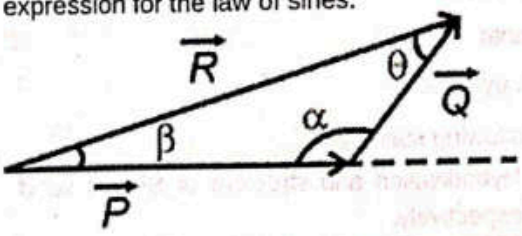
(1) $\vec{A}+\vec{B}=\vec{B}+\vec{A}$

(2) $\vec{A}+\left(\vec{B}+\vec{C}\right)=\left(\vec{A}+\vec{B}\right)+\vec{C}$

(3) $\left|n\vec{A}\right|=n\left|\vec{A}\right|$

(4) All of these

41. For the given arrangement of vectors choose the correct expression for the law of sines.



(1) $\frac{P}{\sin\beta}=\frac{Q}{\sin\alpha}=\frac{R}{\sin\theta}$

(2) $\frac{P}{\sin\beta}=\frac{Q}{\sin\theta}=\frac{R}{\sin\alpha}$

(3) $\frac{P}{\sin\alpha}=\frac{Q}{\sin\theta}=\frac{R}{\sin\beta}$

(4) $\frac{P}{\sin\theta}=\frac{Q}{\sin\beta}=\frac{R}{\sin\alpha}$

42. Choose the vector quantity among the following

(1) Distance

(2) Angular Momentum

(3) Heat

(4) Energy

43. Velocity of particles A and B are $\overrightarrow{v_A}=\left(4\hat{i}+9\hat{j}\right)$ m/sand $\overrightarrow{v_B}=\left(3\hat{i}+2\hat{j}\right)$ m/s. Velocity of B as observed by A is

(1) $-\left(\hat{i}+7\hat{j}\right)\text{ m s}^{-1}$

(2) $\left(\hat{i}+7\hat{j}\right)\text{ m s}^{-1}$

(3) $\left(\hat{i}+3\hat{j}\right)\text{ m s}^{-1}$

(4) $-\left(\hat{i}+3\hat{j}\right)\text{ m s}^{-1}$

44. The magnitude of vector $\hat{i}+3\hat{j}+\sqrt{6}\,\hat{k}$ is

(1) 2 units

(2) 3 units

(3) 4 units

(4) 5 units

45. A car starts from origin at the rate of 5 m/s^2. A man standing on the car fires a bullet at t = 2 s, which hits the bird at t = 8 s having position vector of $\left(70\hat{i}+60\hat{j}+36\hat{k}\right)$ m. The velocity (in m/s) of projection of bullet is [consider the car to be only moving in the positive x-direction and $g=-10\hat{j}\text{ m/s}^2$]

(1) $10\hat{i}+40\hat{j}+6\hat{k}$

(2) $40\hat{i}-30\hat{j}+10\hat{k}$

(3) $30\hat{i}+10\hat{j}+15\hat{k}$

(4) $10\hat{i}+6\hat{j}+40\hat{k}$

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46. The first ionization potential of Na is 5.1 eV. The value of electron gain enthalpy of Na^+ will be

(1) –5.1 eV

(2) –9.1 eV

(3) + 10.2 eV

(4) –2.5 eV

47. Correct order of dipole moment is

(1) $H_2O > HCl > NH_3$

(2) $H_2O > NH_3 > HCl$

(3) $NH_3 > H_2O > HCl$

(4) $HCl > H_2O > NH_3$

48. The pair of elements in which there is a maximum difference of electronegativities on Pauling scale as compared to other given pairs is

(1) K and Br

(2) K and Cl

(3) N and Br

(4) N and F

49. In which of the following compounds expanded octet of central atom is present?

(1) BH_3

(2) H_2O

(3) PCl_5

(4) CCl_4

50. Which among the following has highest lattice enthalpy?

(1) NaI

(2) NaBr

(3) NaCl

(4) NaF

51. Match List-I with List-II.

	List-I		List-II
a.	CH_4	(i)	Polar and planar
b.	XeF_4	(ii)	Polar and non-planar
c.	NH_3	(iii)	Non-polar and planar
d.	H_2S	(iv)	Non-polar and non- planar

Choose the most appropriate answer from the options given below.

(1) a(iv), b(iii), c(i), d(ii)

(2) a(iii), b(iv), c(i), d(ii)

(3) a(iii), b(iv), c(ii), d(i)

(4) a(iv), b(iii), c(ii), d(i)

52. Given below are two statements, one is labelled as assertion (A) and other is labelled as reason (R).

Assertion (A): Maximum covalency shown by boron is four.

Reason (R): Boron does not have any empty orbital in outermost shell at ground state.

In the light of above statements, choose the most appropriate answer from the options given below.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

53. Which of the following is not a d-block element?

(1) $[Kr]4d^1\,5s^2$

(2) $[Xe]\,4f^{14}\,5d^{10}\,6s^2$

(3) $[Xe]4f^2\,5d^0\,6s^2$

(4) $[Xe]\,4f^{14}\,5d^4\,6s^2$

54. The pair of species having same bond order is

(1) O_2, NO^+

(2) O_2^-, N_2

(3) C_2, O_2

(4) O_2^{2-}, N_2

55. Shape of $N(SiH_3)_3$ and $N(CH_3)_3$ will be respectively

(1) Planar, pyramidal

(2) Pyramidal, planar

(3) Planar, planar

(4) Pyramidal, pyramidal

56. Consider the following statements.

Statement I: Hybridisation and structure of SF_4 is sp^3d and see-saw respectively.

Statement II: 180° bond angle is present in ClF_3.

In the light of above statements choose the correct answer.

(1) Statement I and statement II both are true

(2) Statement I is true but statement II is false

(3) Statement I and statement II both are false

(4) Statement I is false but statement II is true

57. Correct order of covalent bond length is

(1) N – O < C – H < C – O

(2) C – H < N – O < C – O

(3) C – H < C – O < N – O

(4) C – O < C – H < N – O

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58. Consider the given statements
(a) He_2 does not exist.
(b) B_2 is paramagnetic in nature.
(c) The magnitude of H-bonding does not depend on the physical state of the compound.
(d) Boiling point of HF is higher than HCl.
The correct statements are

(1) (a) and (b) only

(2) (b) and (c) only

(3) (a), (b) and (d) only

(4) (a), (b), (c) and (d)

59. Which of the following compounds contain ionic, covalent as well as co-ordinate bond?
(I) NH_4Cl
(II) NaOH
(III) $NaBF_4$

(1) (I) only

(2) (I) and (III) only

(3) (II) and (III) only

(4) (I), (II) and (III)

60. Consider the following statements
(a) Bond order of $O-O$ bond in O_3 is 1.5
(b) O_3 is the resonance hybrid of two canonical structures.
(c) Resonance stabilizes the molecule as the energy of resonance hybrid is less than the energy of any single canonical structure.
The correct statements are

(1) (a) and (b) only

(2) (b) and (c) only

(3) (a) and (c) only

(4) (a), (b) and (c)

61. If dipole moment of AB molecule is 1.5 D and A – B bond distance is 160 pm, then the percentage ionic character in AB molecule will be

(1) 25.2%

(2) 19.5%

(3) 31.1%

(4) 42.5%

62. The 1st, 2nd and the 3rd ionization enthalpies I_1, I_2 and I_3 of four atoms with atomic numbers n, n+1, n+2 and n+3, where n < 10, are tabulated below:

Atomic number	Ionization enthalpy (kJ/mol)		
	I_1	I_2	I_3
n	1682	3374	6050
n + 1	2082	3952	6122
n + 2	496	4562	6910
n + 3	739	1461	7733

The value of n is

(1) 7

(2) 8

(3) 9

(4) 6

63. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R)
Assertion (A) : Size of $F^{\ominus}$ ion will be larger than that of F atom
Reason (R) : Addition of one or more electrons in neutral atom would result in increased repulsion among the electrons and a decrease in effective nuclear charge
In the light of above statements, choose the most appropriate answer from the options given below:

(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(2) (A) is correct but (R) is not correct

(3) (A) is not correct but (R) is correct

(4) Both (A) and (R) are correct and (R) is the correct explanation of (A)

64. Consider the following statements:
Statement I: Generally, the lowest energy structure is the one with the smallest formal charges on the atoms.
Statement II: Formal charge on nitrogen atom of NH_4^+ ion is +1.
In the light of above statements, choose the correct option.

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

65. Calcium reacts with element (X) to form an ionic compound. If ground state electronic configuration of (X) is $1s^2 2s^2 2p^5$, then simplest formula of the compound will be

(1) Ca_2X_3

(2) CaX_2

(3) CaX

(4) Ca_2X

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66. Given below are two statements one is labelled as assertion (A) other is labelled as reason (R).
Assertion (A): o-Nitrophenol has higher boiling point than *p*-nitrophenol.
Reason (R) : o-Nitrophenol exhibit intramolecular hydrogen bonding.
In the light of above statements, choose the correct answer from the options given below.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

67. Total number of lone pair(s) of electron on xenon in XeF_2 molecule is

(1) 3

(2) 1

(3) 4

(4) 0

68. The correct order of C — O bond order among CO, CO_3^{2-}, CO_2 is

(1) $CO < CO_3^{2-} < CO_2$

(2) $CO_3^{2-} < CO_2 < CO$

(3) $CO < CO_2 < CO_3^{2-}$

(4) $CO_2 < CO_3^{2-} < CO$

69. Match the species given in column I with the shape given in column II and find the correct option

	Column I		Column II
a.	XeF_4	(i)	Linear
b.	I_3^+	(ii)	Pyramidal
c.	PCl_3	(iii)	Square planar
d.	NO_2^+	(iv)	Bent

(1) a(iv), b(i), c(ii), d(iii)

(2) a(iii), b(i), c(ii), d(iv)

(3) a(iii), b(iv), c(ii), d(i)

(4) a(ii), b(iv), c(i), d(iii)

70. The atom with lowest atomic number that has ground state electronic configuration of $(n-1)d^6\,ns^2$ belongs to

(1) 2nd period

(2) 3rd period

(3) 4th period

(4) 5th period

71. Select the correct order of polarising power of cation.

(1) $Si^{+1} < Al^{+3} < Mg^{+2} < Na^+$

(2) $Na^+ < Mg^{+2} < Al^{+3} < Si^{+4}$

(3) $Na^+ < Mg^{+2} < Si^{+4} < Al^{+3}$

(4) $Al^{+3} < Si^{+4} < Mg^{+2} < Na^+$

72. Identify the incorrect match.

	Name		IUPAC Official Name
(a)	Unnilunium	(i)	Mendelevium
(b)	Unniltrium	(ii)	Lawrencium
(c)	Unnilhexium	(iii)	Seaborgium
(d)	Unununnium	(iv)	Darmstadtium

(1) (b), (ii)

(2) (c), (iii)

(3) (d), (iv)

(4) (a), (i)

73. Consider the following statements
Statement I : The increase in electronegativities across a period is accompanied by an increase in metallic properties.
Statement II : In Pauling scale, an arbitrary value of 4.0 is assigned to electronegativity of fluorine.
In the light of above statements, choose the correct option.

(1) Both statement I and statement II are correct

(2) Statement I is correct but statement II is incorrect

(3) Statement I is incorrect but statement II is correct

(4) Both statement I and statement II are incorrect

74. Consider the given ionization processes
a. $C_2 \longrightarrow C_2^+$
b. $N_2 \longrightarrow N_2^+$
c. $O_2 \longrightarrow O_2^+$
d. $B_2 \longrightarrow B_2^+$
The process(es) for which the bond order gets increased is/are

(1) a and c only

(2) c only

(3) c and d only

(4) b, c and d only

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75. Given below are two statements: one is labelled as Assertion (A) and other is labelled as Reason (R).
Assertion (A): Electrons present in antibonding molecular orbital destabilize the molecule.
Reason (R): Mutual repulsion of the electrons in antibonding molecular orbital is more than the attraction between the electrons and the nuclei which causes a net increase in energy.
In the light of the above statements, choose the correct answer from the options given below.

(1) Both A and R are true and R is the correct explanation of A

(2) Both A and R are true and R is not the correct explanation of A

(3) A is true but R is false

(4) A is false but R is true

76. Correct order of decreasing negative electron gain enthalpy value is

(1) F > Cl > O > S

(2) Cl > F > O > S

(3) S > O > Cl > F

(4) Cl > F > S > O

77. The electronic configuration of an element (X) is $1s^2$, $2s^2$ $2p^6$, $3s^1$. The atomic number of the element which is just below X in the periodic table is

(1) 19

(2) 20

(3) 18

(4) 12

78. Match the oxides given in Column I with their corresponding nature given in Column II and identify the correct code.

Column I	Column II
a. MgO	(i) Acidic
b. CO	(ii) Basic
c. Al_2O_3	(iii) Neutral
d. Cl_2O_7	(iv) Amphoteric

(1) a(i), b(ii), c(iii), d(iv)

(2) a(iv), b(iii), c(ii), d(i)

(3) a(ii), b(iii), c(iv), d(i)

(4) a(ii), b(iii), c(i), d(iv)

79. Which among the following is not a representative element?

(1) Rb

(2) Fe

(3) In

(4) Te

80. $d\pi - p\pi$ bond present in

(1) CO_3^{2-}

(2) PO_4^{3-}

(3) NO_3^-

(4) NO_2^-

81. Pick up a pair of elements showing diagonal relationship with each other

(1) Na and K

(2) Mg and Al

(3) Li and Mg

(4) Be and Si

82. The hybridisation of central atom of the molecules NH_3, SO_3 and CO_2 respectively are

(1) sp^2, sp^3 and sp

(2) sp^3, sp^2 and sp

(3) sp^3, sp^3 and sp^2

(4) sp^2, sp^3 and sp^3

83. Given below are the two statements:
Statement I: Bond length is defined as the equilibrium distance between the nuclei of two bonded atoms in a molecule.
Statement II: The covalent radius represents the overall size of the atom which includes its valence shell in a nonbonded situation.
In the light of above statements, choose the correct answer from options given below.

(1) Statement I is correct but statement II is incorrect

(2) Both statement I and statement II are correct

(3) Both statement I and statement II are incorrect

(4) Statement I is incorrect but statement II is correct

84. During conversion of O_2 to O_2^-, coming electron will enter in

(1) $\pi 2p_x$

(2) $\sigma 2p_z$

(3) $\pi^* 2p_x$

(4) $\sigma^* 2s$

85. Consider the following statements:
Statement I: XeF_6 has distorted octahedral shape.
Statement II: Total number of bond pairs around central atom in XeF_6 are 6.
In the light of above statements, choose the **correct** answer from options given below.

(1) Both statements I and II are correct

(2) Statement I is incorrect but statement II is correct

(3) Statement I is correct but statement II is incorrect

(4) Both statements I and II are incorrect

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86. Which of the following has the highest percentage of ionic character?

(1) $BeCl_2$

(2) $MgCl_2$

(3) $CaCl_2$

(4) $BaCl_2$

87. Consider the following statements.
(i) Bromine is more electronegative as compared to sulphur (on Pauling scale).
(ii) Both Barium and Boron are main group elements.
(iii) Tin and lead are metals of *p*-block.
(iv) The number of unpaired electrons in *d* subshell of iron is 6.
The incorrect statements is/are

(1) (i) and (iv) only

(2) (iv) only

(3) (ii) only

(4) (i), (ii), (iii) and (iv)

88. First and second I.P of Mg are 737.76 and 1450.73 kJ/mole respectively. The energy required to convert all the atoms of magnesium to Mg^{2+} ions present in 24 g of magnesium vapours is

(1) 2188.5 kJ

(2) 1094.7 kJ

(3) 104.2 kJ

(4) 595.5 kJ

89. The element with atomic number 58 belongs to

(1) *s*-block

(2) *p*-block

(3) *d*-block

(4) *f*-block

90. Consider the following statements
(a) Cerium (Ce) and Lawrencium (Lr) are lanthanoids
(b) Zn and Cd are transition elements
(c) Americium (Am) and Curium (Cm) are transuranium elements
(d) IUPAC symbol of Une is Mt
The correct statements are

(1) (c) and (d) only

(2) (a) and (c) only

(3) (b) and (d) only

(4) (a), (b) and (d) only

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91. State **true (T)** or **false (F)** for the given statements and select the **correct** option.
A) After meiosis I, the resultant daughter cells have twice the number of chromatids as well as the amount of DNA in comparison to haploid gametes.
B) Reorganisation of all cell components take place during the G_1 phase.
C) The synapsed homologous chromosomes, become clearly visible as tetrad at pachytene stage.
D) Anaphase and anaphase II are similar in a single individual in terms of the genetic variability of chromatids.

	A	B	C	D
(1)	T	F	T	F
(2)	F	T	F	T
(3)	T	T	F	F
(4)	T	F	F	T

(1) (1)
(2) (2)
(3) (3)
(4) (4)

92. Match the following columns and choose the **correct** option.

	Column I		Column II
A.	Diakinesis	(i)	It can last for months or years in oocytes of some vertebrates
B.	Pachytene	(ii)	The nucleolus disappear and the nuclear envelope also breaks down
C.	Diplotene	(iii)	Formation of synaptonemal complex occurs
D.	Zygotene	(iv)	Recombination between homologous chromosomes is completed by the end of this phase

(1) A(ii), B(i), C(iv), D(iii)
(2) A(ii), B(iv), C(i), D(iii)
(3) A(ii), B(iv), C(iii), D(i)
(4) A(iii), B(iv), C(i), D(ii)

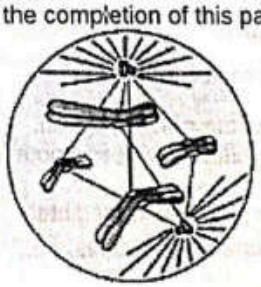
93. Interkinesis and interphase both are similar, as both
a. Are the phase, preparing the cell for upcoming cell division.
b. Have extremely extended form of chromosome that is composed of DNA, RNA, histone and non-histone proteins.
c. Show duplication of the centrioles in the cytoplasm of animal cell.
d. Are associated with the synthesis of RNA and proteins.
The **correct** ones are:

(1) only b and d
(2) only a and d
(3) only c and d
(4) only a, c and d

94. Observe the following events:
a. Formation of the site of active ribosomal RNA synthesis.
b. Formation of the cell plate in the centre to form two daughter cells.
c. Complete disappearance of perinuclear space.
d. Centrosome moving towards the pole.
e. Attachment of spindle fibres to the sites found on the primary constriction.
f. Splitting of primary constriction.
Select the option depicting the **correct** sequence of events that occur during the formation of liquid endosperm in coconut.

(1) d → c → e → f → a → b
(2) c → e → f → d → b
(3) e → f → d → c → a
(4) c → e → f → a

95. The following diagram is representing a transition stage to a particular phase of cells dividing mitotically. One of the characteristic feature of the phase which occurs just after the completion of this particular phase is



(1) Condensation of chromatin material
(2) Movement of centrioles to the opposite poles
(3) Assembling of nuclear envelope
(4) Splitting of centromeres

96. Match the column I with column II.

	Column I		Column II
a.	Bivalent	(i)	Multinucleated condition in some organisms
b.	Chiasmata	(ii)	More clearly visible at the third sub-stage of prophase I
c.	Recombination nodules	(iii)	Becomes visible during the fourth sub-stage of prophase I
d.	Syncytium	(iv)	Facilitate an enzyme mediated process, completed by the end of pachytene

Choose the **correct** option.

(1) a(ii), b(iv), c(iii), d(i)

(2) a(iii), b(iv), c(ii), d(i)

(3) a(iv), b(ii), c(i), d(iii)

(4) a(ii), b(iii), c(iv), d(i)

97. The stage between meiotic I and meiotic II divisions is called __A__ and is generally __B__.

	A	B
(1)	Interkinesis	Long-lived
(2)	Interphase	Long-lived
(3)	Interphase	Short-lived
(4)	Interkinesis	Short-lived

(1) (1)

(2) (2)

(3) (3)

(4) (4)

98. During resting phase of cell cycle

(1) DNA does not replicate

(2) Cell organelles duplicate

(3) Cell becomes metabolically inactive

(4) Chromatids condense to form chromosomes

99. Read the given statements and select the **correct** option.
Statement A : The plants can show mitotic division in both haploid and diploid cells.
Statement B : The G_1 phase of interphase is immediately followed by the first stage of karyokinesis of mitosis, i.e., Prophase.

(1) Only statement A is correct

(2) Only statement B is correct

(3) Both statements are correct

(4) Both statements are incorrect

100. During prophase

(1) Chromosomes lie at equator of cell

(2) Initiation of condensation of chromosomal material occurs

(3) Chromosomes migrate towards opposite poles

(4) Centromere splits

101. All of the following are significance of mitosis, **except**

(1) It produces genetic variations during sexual reproduction.

(2) It is essential for growth and development of an organism.

(3) It helps in healing of wounds

(4) It helps to restore the nucleo-cytoplasmic ratio

102. Prophase-II of meiotic division

(1) Is preceded by centrioles duplication in animals

(2) Is followed by DNA duplication in plants

(3) Begins with the simultaneous splitting of the centromere of each chromosome

(4) Is marked by reappearance of nucleolus

103. A bivalent of prophase I is referred as

(1) One centromere with one chromatid

(2) Two centromeres with two chromatids

(3) Two centromeres with four chromatids

(4) Four centromeres with four chromatids

104. Read the given statements and choose the **correct** option.
Statement A: The final stage of prophase I is marked by dissolution of synaptonemal complex and can last for years in some vertebrates.
Statement B: During mitosis, the best stage to study morphology of chromosomes is characterized by the event in which all the chromosomes come to lie at the equator.

(1) Only statement A is correct

(2) Only statement B is correct

(3) Both statements A and B are correct

(4) Both statements A and B are incorrect

105. "The two __A__ together with spindle fibres forms __B__."
Complete the above statement by choosing the **correct** option for A and B.

	A	B
(1)	Microtubules	Mitotic apparatus
(2)	Asters	Mitotic apparatus
(3)	Microtubules	Centriole
(4)	Asters	Centriole

(1) (1)

(2) (2)

(3) (3)

(4) (4)

106. In most of the animals, mitotic cell division is seen in

(1) Oocytes

(2) Diploid somatic cells

(3) Gametes

(4) Haploid germ cells

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107. The mechanism by which conservation of specific chromosome number of each species is achieved across generations in sexually reproducing organisms is

(1) Mitosis

(2) Meiosis

(3) Interphase

(4) Cytokinesis

108. Choose **incorrect** option w.r.t. cell cycle

(1) Cell growth in terms of cytoplasmic increase occurs continuously

(2) DNA synthesis occurs during specific stage

(3) Events are not under genetic control

(4) Cell duplicates its genome once

109. Nucleolus, Golgi complex and ER reappear during

(1) Prophase

(2) Metaphase

(3) Telophase

(4) Anaphase

110. Best stage to study the shape of chromosomes is

(1) Anaphase

(2) Metaphase

(3) Prophase

(4) Telophase

111. Bivalent chromosomes align themselves at the equator during

(1) Metaphase I

(2) Prophase I

(3) Metaphase II

(4) Anaphase II

112. Shortening of microtubules attached to the kinetochores takes place during

(1) Third stage of karyokinesis in equational and reductional division.

(2) Second and third substage of prophase I.

(3) Final stage of M-phase.

(4) Fourth stage of meiosis and mitosis.

113. Among eukaryotes, chromosome duplication occurs during the

(1) G_1 phase

(2) S phase

(3) G_2 phase

(4) G_0 phase

114. The longest phase of cell cycle is

(1) Metaphase

(2) Prophase

(3) Interphase

(4) Anaphase

115. Spindle fibres attach to kinetochores of chromosomes during

(1) Early prophase

(2) Anaphase

(3) Metaphase

(4) Telophase

116. Select the events which occur twice during meiosis.

a. Nuclear division
b. Division of cytoplasm
c. DNA replication
d. Bivalent formation
e. Equational division

(1) a and b

(2) a, b and e

(3) a, b and d

(4) c, d and e

117. If a meiocyte contains 20 chromosomes, then what will be the chromosome number at each pole during anaphase-I?

(1) 40

(2) 20

(3) 10

(4) 5

118. Observe the following taxonomical categories.

a. *Musca*
b. Insecta
c. Muscidae
d. Arthropoda
e. Diptera

Which of the following options depicts the correct ascending order regarding taxonomical categories for a housefly?

(1) a → d → b → e → c

(2) b → e → c → d → a

(3) a → c → e → b → d

(4) c → a → e → b → d

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119. Read the following statements carefully and select option for the set of **correct** statements.
(a) The classical taxonomy is based on observable morphological characters.
(b) In all unicellular organisms, growth and reproduction are mutually exclusive events.
(c) Mayr was awarded the three prizes widely regarded as the triple crown of biology
(d) Mountains, boulders and sand mounds do not grow.

(1) (a) & (b)

(2) (a) & (c)

(3) (b) & (c)

(4) (b) & (d)

120. Read the following statements and select the correct option.
Statement A: Taxonomic categories are distinct biological entities and not merely morphological aggregates.
Statement B: Classification of organisms into various categories does not help in identifying similarities and dissimilarities among different organisms.

(1) Only A is incorrect

(2) Only B is incorrect

(3) Both A and B are correct

(4) Both A and B are incorrect

121. What do A, B and C represent in the given scientific epithet respectively?

Mangifera	*indica*	Linn
C	B	A

(1) Generic name, specific name and author's name

(2) Specific name, generic name and author's name

(3) Author's name, specific epithet and generic name

(4) Generic name, author's name and specific name

122. *Solanum*, *Petunia* and *Datura* are three different ___A___ and they are placed in family ___B___.
Select the correct option for A and B.

(1) A – Genera; B – Solanaceae

(2) A – Species; B – Solanaceae

(3) A – Families; B – Convolvulaceae

(4) A – Families, B – Poaceae

123. When we move from species to kingdom, the number of common characteristics

(1) Goes on increasing

(2) Remains same

(3) Goes on decreasing

(4) Increases but gradually decreases after class

24. Mark the **incorrect** match w.r.t taxonomic categories of mango.

(1) Kingdom – Angiospermae

(2) Order – Sapindales

(3) Family – Anacardiaceae

(4) Genus – *Mangifera*

125. Mark the taxon that is equivalent to the taxon 'insecta' of house fly, in taxonomical hierarchy.

(1) Poaceae of wheat

(2) Hominidae of man

(3) Dicotyledonae of brinjal

(4) Sapindales of mango

126. Systematics

(1) Is derived from a Greek word

(2) Does not include classification of organisms

(3) Is considered as taxonomy along with phylogeny

(4) Does not deal with diversity of organisms

127. The taxon which has less similar characters than the members of a class is

(1) Order

(2) Family

(3) Division

(4) Genus

128. A recently discovered organism was given a scientific name according to the rules assigned in ICZN. Which of the following features should **not** be related to this organism?

(1) Capacity to convert glucose into starch

(2) Presence of semi-autonomous cell organelle

(3) Ability to detect changes in the environment

(4) Plasma membrane as the outermost covering of the cell

129. Fill in the blanks A and B.
Kingdom ⇢ Phylum ⇢ [A] ⇢ Order ⇢ [B]

(1) A - Genus; B - Species

(2) A - Family; B - Class

(3) A - Class; B - Family

(4) A - Species; B - Division

130. Who proposed most accepted "Biological concept of species"?

(1) Lamarck

(2) Ernst Mayr

(3) Julian Huxley

(4) Carolus Linnaeus

131. Out of the following taxa which one represents the highest rank in taxonomic hierarchy?

(1) Poales

(2) Anacardiaceae

(3) Monocotyledonae

(4) Polymoniales

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132. The branch of science which deals with the study of principles and procedures of classification is known as

(1) Identification

(2) Taxonomy

(3) Nomenclature

(4) Ecology

133. According to the rules of binomial nomenclature of organisms

(1) Both generic name and specific epithet are printed in italics

(2) Genus name starts with small letter

(3) Only specific epithet is printed in italics

(4) Specific epithet is written before genus

134. Cats and dogs are respectively placed in the families

(1) Felidae and Hominidae

(2) Muscidae and Felidae

(3) Poaceae and Canidae

(4) Felidae and Canidae

135. According to IUCN, currently known and described species of all organisms ranges between

(1) 1.7 – 1.8 billion

(2) 1.7 – 1.8 million

(3) 7 – 8 million

(4) 17 – 18 billion

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Zoology

136. Rate of breathing is increased by

(1) Increased acidity of blood

(2) Increased alkalinity of blood

(3) High levels of oxygen in blood

(4) Low temperature of body

137. **Assertion (A) :** Abdominal muscles play an important role during quiet inspiration.
Reason (R) : They contract so that volume of pulmonary cavity increases for air to pull in from outside.
In the light of the above statements, choose the correct option.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

138. What is the function of respiratory part of human respiratory system?

(1) It clears the incoming air from foreign particles

(2) It brings the temperature of air upto the body temperature

(3) It transports the atmospheric air

(4) It exchanges O_2 and CO_2 between blood and atmospheric air

139. The toxic effect of carbon monoxide is due to its greater affinity for haemoglobin as compared to oxygen, approximately by

(1) 2 times

(2) 20 times

(3) 200 times

(4) 1000 times

140. Which of the following conditions will facilitate association of oxygen with haemoglobin?

(1) Increase in temperature

(2) Decrease in pCO_2

(3) Increase in BPG

(4) Increase in H^+ ions

141. Tidal volume in a human is defined as

(1) The total maximum air that can be driven into the lungs

(2) The volume of inspired or expired air during a normal breathing

(3) The air that is left in the lungs after normal expiration

(4) The air that can be expelled out forcibly after the normal expiration

142. During expiration in a resting human, which one of the following changes occurs?

(1) The epiglottis actively closes glottis to prevent air from escaping through the trachea

(2) Residual lung volume reduces due to enhance air expulsion

(3) Muscle fibres of diaphragm undergo active contraction, increasing intrapulmonary pressure

(4) Elastic recoiling of the lungs leads to increase in intrapulmonary pressure

143. Select the **incorrect** statement.

(1) Inflammation of bronchi and bronchioles is seen in asthma

(2) One of the major causes of Emphysema is cigarette smoking

(3) Long exposure to large amount of dust could lead to fibrosis of lungs

(4) Exchange of gases increases in emphysema due to inflation of lungs

144. Select the **correct** statement.

(1) Trachea divides into right and left bronchioles at the level of 5th thoracic vertebra

(2) Surfactants prevent collapsing of lung alveoli

(3) Larynx is common passage for both food and air

(4) Some amount of respiratory gas exchange takes place in bronchi from atmospheric air as it is well supplied with blood vessels

145. Choose the **incorrect** match w.r.t. respiratory volumes.

(1) FRC = ERV + RV

(2) EC = TV + ERV

(3) VC = ERV + TV + IRV

(4) TLC = ERV + RV + IRV

146. A chemosensitive area situated adjacent to the respiratory rhythm centre is highly sensitive to

(1) CO_2 content in arterial blood

(2) Acidic pH in venous blood

(3) $[H^+]$ concentration in venous blood

(4) pO_2 in arterial blood

147. What percentage of O_2 and CO_2 respectively are carried in dissolved state through the plasma during their transport in humans?

(1) 7 and 3

(2) 3 and 7

(3) 97 and 70

(4) 70 and 97

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148. Thoracic chamber in humans is formed by

	Dorsal side	Ventral side	Lateral side	Lower side
(1)	Diaphragm	Ribs	Vertebral column	Sternum
(2)	Vertebral column	Sternum	Diaphragm	Ribs
(3)	Vertebral column	Sternum	Ribs	Diaphragm
(4)	Sternum	Ribs	Vertebral column	Diaphragm

(1) (1)

(2) (2)

(3) (3)

(4) (4)

149. Consider the given statements.

Statement A: Cellular respiration deals with utilisation of O_2 by the cells for catabolic reactions and resultant release of CO_2.

Statement B: In pulmonary ventilation, atmospheric air is drawn in and CO_2 rich alveolar air is released out.

In the light of above statements, choose the correct option.

(1) Both statements A and B are correct

(2) Both statements A and B are incorrect

(3) Only statement A is correct

(4) Only statement B is correct

150. Which of the following changes usually tend to occur in the plain dwellers when they move to high altitudes?

(i) Increased number of RBCs per unit volume
(ii) Decrease in the size of RBCs
(iii) Decrease in heart rate
(iv) Increase in breathing rate
Choose the **correct** option.

(1) (i), (ii) and (iv) only

(2) (i) and (iv) only

(3) (i), (ii), (iii) and (iv)

(4) (i) and (ii) only

151. Match column I with column II and choose the **correct** one from the given options.

	Column I Animals		Column II Respiratory organs
a.	Lizard	(i)	Gills
b.	Earthworm	(ii)	Tracheal tube
c.	Pearl oyster	(iii)	Lungs
d.	Honey Bee	(iv)	Moist cuticle

(1) a(i), b(ii), c(iii), d(iv)

(2) a(iv), b(ii), c(iii), d(i)

(3) a(iii), b(iv), c(i), d(ii)

(4) a(ii), b(iv), c(iii), d(i)

152. One molecule of haemoglobin can bind maximally with

(1) One molecule of CO_2

(2) Two molecules of O_2

(3) Eight molecules of O_2

(4) Four molecules of O_2

153. Given below are the steps involved in human respiration.

(i) Drawing in of atmospheric air by pulmonary ventilation.
(ii) Transport of gases by blood.
(iii) Diffusion of O_2 and CO_2 between blood and tissue.
(iv) Diffusion of respiratory gases across the diffusion membrane.
(v) Utilization of O_2 by the cells for catabolic reactions.

Select the **correct** sequence in which the above mentioned steps take place.

(1) (i) → (ii) → (iii) → (iv) → (v)

(2) (i) → (iv) → (ii) → (iii) → (v)

(3) (ii) → (iii) → (i) → (iv) → (v)

(4) (i) → (iii) → (ii) → (iv) → (v)

154. Match the items given in column I with those given in column II and choose the **correct** option.

	Column I (pO_2 and pCO_2 at different body parts)		Column II (in mm Hg)
a.	pO_2 in lung alveoli	(i)	40
b.	pO_2 in systemic veins	(ii)	45
c.	pO_2 in systemic arteries	(iii)	95
d.	pCO_2 in body tissues	(iv)	104

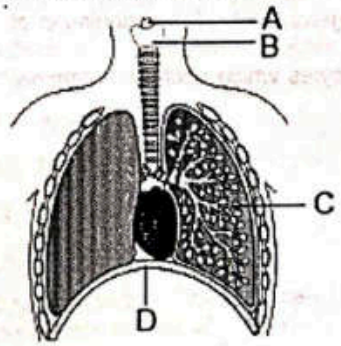
(1) a(i), b(ii), c(iii), d(iv)

(2) a(iv), b(ii), c(iii), d(i)

(3) a(iv), b(i), c(iii), d(ii)

(4) a(iii), b(i), c(iv), d(ii)

155. Given below is a diagrammatic view of the human respiratory system with labels A, B, C and D. Choose the option with **correct** identification and main function/characteristic.



(1) A - Glottis - Prevents entry of food into the larynx during swallowing

(2) D - Diaphragm - Relaxes and becomes flat during inspiration

(3) B - Trachea - Leads into larynx

(4) C - Alveoli - Thin walled bag-like structures for exchange of gases

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156. Select the correct statement w.r.t. the nodal tissues of human heart.

(1) Entire heart is made up of specialized tissue called nodal tissue.

(2) Nodal musculature has the ability to generate action potentials only due to the external stimuli.

(3) Both sino-atrial and atrio-ventricular nodes are located in left atrium.

(4) Number of action potentials that could be generated in a minute vary at different parts of nodal tissue.

157. Complete the analogy and choose the **correct** option.
Angina : Acute chest pain :: Congestive heart failure : _____

(1) Heart stops beating

(2) Affect the vessels that supply to heart muscles

(3) Heart muscles are suddenly damaged

(4) Congestion of the lungs

158. During isovolumetric ventricular systole,

(1) Pressure within the ventricles decreases

(2) Volume of blood within the ventricles decreases

(3) Volume of blood remains unchanged in ventricles

(4) AV valves remain open

159. The function of chordae tendinae is to

(1) Close the mitral valves during atrial systole

(2) Close the semilunar valves during ventricular systole

(3) Close the atrio-ventricular valves during joint diastole

(4) Prevent AV-valves from averting into atria during ventricular systole

160. Consider the following four statements and identify them as true(T) or false(F).
A – Neutrophils are agranulocytes, that constitute 60–65% of total WBCs and are phagocytic cells.
B – Eosinophils(2–3%of total WBCs) are granulocytes, that resist infections and are also associated with allergic reactions.
C – Platelets are produced from megakaryocytes of bone marrow that secrete fibrinogens involved in coagulation of blood.
D – Basophils are granulocytes which secrete histamine, serotonin and heparin.
Choose the correct option.

	A	B	C	D
(1)	T	T	F	T
(2)	F	T	F	T
(3)	F	F	F	T
(4)	T	F	T	F

(1) (1)

(2) (2)

(3) (3)

(4) (4)

161. Arteries are different from veins in having

(1) Thick and muscular walls

(2) Valves

(3) Collapsible wall when empty

(4) Slow blood flow

162. Choose the **incorrect** statement.

(1) Heart is an endodermally derived organ in humans

(2) Human blood vascular system consists of muscular chambered heart

(3) Humans have a closed circulatory system

(4) Humans exhibit complete double circulation

163. Choose the **incorrect** statement w.r.t. tissue fluid.

(1) It is also called interstitial fluid

(2) This fluid is drained in lymphatic vessels where it is called lymph

(3) It is rich in nutrients but lacks respiratory gases.

(4) Its mineral content is same as that of plasma.

164. Parameters that change during strenuous exercise in humans, is/are
a. Cardiac output
b. Stroke volume
c. Heart rate
d. Duration of cardiac cycle

(1) a and b only

(2) c and d only

(3) a, b and c only

(4) a, b, c and d

165. Given below are two statements w.r.t. a human. One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A): Hemoglobin is enclosed in red blood cells and therefore not excreted in urine in normal conditions.
Reason (R): Haemoglobin is a metaloprotein present in humans.
In the light of above statements, select the correct option.

(1) Both Assertion and Reason are true and the Reason is the correct explanation of the Assertion

(2) Both Assertion and Reason are true but the Reason is not the correct explanation of the Assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

166. Read the following statements and choose the correct option.
Statement A: In humans, under normal physiological conditions, each ventricle pumps out about 5000 mL blood during each cardiac cycle.
Statement B: Each cardiac cycle involves systole and diastole of all the four chambers of the heart.

(1) Both statements A and B are correct.

(2) Both statements A and B are incorrect

(3) Only statements A is incorrect

(4) Only statement B is incorrect

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167. Mammals are said to have a "double circulatory system". This means

(1) That the blood vessels are paired

(2) That there are two types of blood vessels attached to every organ; an artery and a vein

(3) That there are two system; one from the heart to the lungs and back to the heart and other to and from heart to rest of the body

(4) That the blood circulates twice as quickly through left ventricle during each cardiac cycle

168. How many biomolecules/ions mentioned in the box below are present in the blood plasma?

Cl^-, lipids, Mg^{++}, Na^+, glucose, amino acids

Choose the **correct** option.

(1) Three

(2) Four

(3) Five

(4) Six

169. Consider the following structures w.r.t. humans

(a) Left ventricle
(b) Right ventricle
(c) Aorta
(d) Pulmonary artery

In which of the above mentioned structures, the epithelial lining comes in direct contact with the oxygenated blood?

(1) (a) and (c)

(2) (a) and (d)

(3) (b) and (c)

(4) (b) and (d)

170. Which of the following wave in ECG represents depolarisation of the atria?

(1) P-wave

(2) Q-wave

(3) QRS wave

(4) T-wave

171. Which one of the following statements is **correct**?

(1) Blood moves freely from atrium to the ventricle during joint diastole.

(2) Increased ventricular pressure causes closing of the semilunar valves.

(3) The atrio-ventricular node (AVN) generates an action potential to stimulate atrial contraction, under normal conditions.

(4) The tricuspid and the bicuspid valves open due to the pressure exerted by the simultaneous contraction of the atria

172. How many double circulations are normally completed by the human heart in 30 seconds?

(1) Eight

(2) Sixteen

(3) Seventy two

(4) Thirty six

173. Red blood cells of a person 'X' do not agglutinate with addition of either anti-A or anti-B serum. However, serum collected from his blood causes agglutination of RBCs collected from another person. 'Y' which has no antibodies against blood group antigens

Based on the above information, blood group of the person 'X' is most likely.

(1) A

(2) B

(3) O

(4) AB

174. In human heart, a thin muscular wall separates

(1) Right and left ventricles

(2) Right auricle from right ventricle

(3) Right and left atria

(4) Left auricle from left ventricle

175. Which of the following is an **incorrect** match?

(1) Heart sound Lub – Closure of AV valves during ventricular systole

(2) Heart sound Dub – Opening of semilunar valves during ventricular diastole

(3) Cardiac output – (Stroke volume) × (Heart rate)

(4) Incomplete double circulation – Frog

176. Select the **correct** option related to the regulation of cardiac activity.

(1) Human heart is neurogenic in nature

(2) Centres in pons initiate heart beat

(3) Parasympathetic signals tend to decrease heart rate

(4) Adrenal medullary hormones decrease heart rate

177. Choose the **odd** one w.r.t. open circulatory system.

(1) Most annelids

(2) Arthropods

(3) Hemichordates

(4) Non-cephalopod molluscs

178. Which of the following globulins in blood participate in immune reactions?

(1) Alpha globulins

(2) Beta globulins

(3) Gamma globulins

(4) Myoglobin

179. Erythroblastosis foetalis can occur in a subsequent pregnancy in case of

(1) Rh+ve mother with Rh+ve foetus

(2) Rh–ve mother with Rh+ve foetus

(3) Rh-ve mother with Rh-ve foetus

(4) Rh+ve mother with Rh–ve foetus

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180. Match the column I and II. Choose the **correct** option.

	Column I		Column II
a.	Plasma cells	(i)	Secretion can constrict blood vessels
b.	Mast cells	(ii)	Ingest foreign matter
c.	Macrophages	(iii)	Produce antibodies

(1) a(i), b(ii), c(iii)

(2) a(ii), b(iii), c(i)

(3) a(iii), b(i), c(ii)

(4) a(ii), b(i), c(iii)

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